Quartic-phase perturbations of core energy in Gaussian and caustic beams

Auto BeamScientist Project

Abstract

A radial quartic phase changes the local wavefront curvature across an optical beam. How this redistribution affects energy collected by a fixed aperture depends on both the input field and the propagation distance. We compare Gaussian, Pearcey and Butterfly fields with and without the same quartic phase in free space, using the median energy fraction within an on-axis circular core over a fixed axial interval. At a phase coefficient of $a_4 = 0.7$, this fraction decreases by 1.60%, 2.21% and 1.66%, respectively. The Gaussian and Pearcey changes retain their sign under mesh refinement, a larger computational window and an independent Fresnel calculation. A Gaussian chirp-plus-ripple comparison produces a larger decrease of 29.0%. These observations characterize a similar direction of response in the three fields at the selected coefficient; they do not establish equal responses across beam families. The separate hypothesis that an intensity-moment-matched quadratic phase reproduces the quartic response remains open because that comparison was not simulated.

1 Introduction

A phase mask can redistribute optical energy without changing the power at the input plane. The resulting intensity pattern is often described through its peak or its focal position. For a finite collection aperture, another quantity matters: how much of the incident power remains near the optical axis over the propagation interval of interest. A bright local maximum and a large collected energy fraction are different properties, particularly for fields with extended lobes or curved propagation trajectories.

Pearcey, Butterfly and finite-energy Airy fields offer different transverse structures for examining this distinction [5, 11, 1, 2]. Here we apply a common radial quartic phase to these fields and to a Gaussian reference. Its phase gradient grows cubically with radius, whereas that of a quadratic phase grows linearly. The two transformations can therefore redistribute energy differently even when their coefficients are matched using an intensity moment. Whether a single moment-based coefficient captures the energy collected by a small on-axis core is a concrete question that wave propagation can test.

We study the first part of that comparison: the change caused by the quartic phase relative to each unmodified input. The measurement is fixed before propagation and uses the same power, aperture and axial interval for every field. The Gaussian, Pearcey and Butterfly cases all exhibit a small decrease in core energy at the selected coefficient. Additional calculations examine how the Gaussian and Pearcey changes vary with numerical sampling. A separate quadratic surrogate was proposed but was not included in the simulations, so the results characterize the quartic response rather than resolve the proposed moment equivalence.

2 Related Work

The Pearcey field provides a cusp-caustic construction with autofocusing and self-healing behavior [5]. Symmetric Pearcey-Gaussian beams and masked Pearcey arrays use related constructions to alter propagation and focusing [9, 8]. Butterfly fields extend this setting to higher-order optical

catastrophes [11], while finite-energy Airy fields provide a familiar example of curved intensity trajectories [1, 2]. Propagation through an anisotropic medium adds another source of structural change, as studied for Pearcey beams in uniaxial crystals [10]. The present comparison keeps the environment as free space and varies the applied phase, with a fixed on-axis energy measure throughout.

Phase-only reshaping has also been investigated through beam-quality measures such as M^2 [7]. A circular-core energy fraction probes a different aspect of the propagated field: energy outside the chosen radius contributes no signal even when it forms an intense off-axis feature. Our propagation calculations use an angular-spectrum solver and a direct Fresnel comparison. Matsushima and Shimobaba describe a band-limited angular-spectrum construction [4]; the solver used here instead applies a conservative radial spectral cutoff, as specified below.

3 Theoretical model

3.1 Input fields and quartic phase

Each field is sampled on a Cartesian grid and normalized to the same input power P:

$$U_0(x_j, y_k) = \sqrt{\frac{P}{\Delta x^2 \sum_{a,b} |V_{ab}|^2}}\, V(x_j, y_k), \tag{1}$$

where Δx is the grid spacing. The fields are truncated by a circular aperture before normalization. The Gaussian reference has amplitude $V = \exp(-r^2/w^2)$, with $r^2 = x^2 + y^2$ and $w = 300\,\mu$m.

The finite-energy Airy input is

$$V(x, y) = \mathrm{Ai}\Big(\frac{x}{s}\Big)\,\mathrm{Ai}\Big(\frac{y}{s}\Big)\exp\left[\alpha\frac{x+y}{s}\right], \tag{2}$$

with scale s and exponential apodization α. The Pearcey and Butterfly inputs use Gaussian-apodized catastrophe integrals,

$$V(x, y) = C_n\Big(\frac{x}{s}, \frac{y}{s}\Big)\exp\left(-\frac{r^2}{w^2}\right), \qquad C_n(u, v) = \int_{-\infty}^{\infty} \exp\big[i\left(t^n + vt^2 + ut\right)\big]\, dt, \tag{3}$$

where $n = 4$ gives the Pearcey field and $n = 6$ gives the Butterfly slice used here. The additional Butterfly control parameters are zero. The integrals are evaluated by contour rotation with 192 quadrature points; a 384-point calculation checks their discretization. In these definitions, s sets the transverse catastrophe scale and w sets the Gaussian envelope. The parameter α belongs to the Airy field alone.

The applied quartic phase is

$$U_q(x, y) = U_0(x, y)\exp\left[ia_4\left(\frac{r}{w}\right)^4\right]. \tag{4}$$

All primary quartic comparisons use $a_4 = 0.7$. A separate Gaussian comparison uses the phase $a_2(r/w)^2 + b\sin[2(r/w)^2]$, with $a_2 = 0.7$ and $b = 0.5$. This chirp-plus-ripple mask is radial and deterministic. Its coefficients were specified directly; they were not obtained from a moment-matching calculation.

3.2 Propagation and energy measurement

For either launch field $U_{\mathrm{in}} = U_0$ or U_q, propagation is computed as

$$U_z = \mathcal{F}^{-1}\Big[\mathcal{F}(U_{\mathrm{in}})\, B_z\, \exp\Big(i2\pi z\sqrt{\lambda^{-2} - f_x^2 - f_y^2}\Big)\Big], \tag{5}$$

where λ is the wavelength and B_z is the solver's radial spectral cutoff. This filter removes evanescent and high-spatial-frequency components; it is distinct from the full band-limited construction of Ref. [4]. The independent check evaluates the paraxial Fresnel diffraction integral in real space using separable matrix quadrature.

The measured quantity is the fraction of input power inside a circular core,

$$C(z) = P^{-1} \int_{x^2+y^2 \le r_c^2} |U_z|^2 \, dA, \qquad \overline{C} = \mathrm{median}_{z \in \mathcal{Z}} \, C(z), \tag{6}$$

with $r_c = 50\,\mu$m. We write $\eta = \overline{C}$ and report the relative change $(\eta_q - \eta_0)/\eta_0$. The core remains centered on the optical axis; it does not follow a beam's intensity maximum or centroid. The same radius and axial planes are used before and after the phase transformation.

The calculation also records normalized coherent copies,

$$U_c = \sqrt{\frac{P}{\int |U + a e^{i\delta} \mathcal{R}_\theta U|^2 dA}} \left(U + a e^{i\delta} \mathcal{R}_\theta U \right), \tag{7}$$

where $\mathcal{R}_\theta$ rotates the transverse field. Here $\theta = 0$, $\delta = 0$ and $a = 1$, so Eq. (7) reproduces the normalized single field for every input. These copies are redundant measurements, rather than a parity-selection comparison of the kind used in spatial-mode sorting [3].

3.3 Proposed quadratic surrogate

The original research question proposed replacing the quartic phase by a quadratic phase with coefficient

$$M = \frac{\sum_{j,k} |U_{0,jk}|^2 (r_{jk}/w)^4}{\sum_{j,k} |U_{0,jk}|^2 (r_{jk}/w)^2}, \qquad q_{2,\mathrm{eff}} = a_4 M. \tag{8}$$

This rule assigns a coefficient from intensity moments. Equality of coefficients under such a rule is not an identity between the propagated fields or their core energies. A quartic phase also changes a Gaussian-amplitude input into a non-Gaussian complex field, whose propagation is not described by a single Gaussian q parameter.

The proposed comparison statistic was

$$R = \frac{|\eta_{\mathrm{quartic}} - \eta_{\mathrm{equiv}}|}{|\eta_{\mathrm{equiv}} - \eta_{\mathrm{base}}|}, \tag{9}$$

for a nonzero denominator. The planned bands were $R \le 0.15$ for the Gaussian reference, $R > 0.25$ for a caustic residual, and $R \le 0.10$ for a small residual. These thresholds concern R, not the quartic-versus-base percentage change. The equivalent-quadratic fields and their moment coefficients were not calculated, so R is unavailable in the present results.

4 Numerical methods

The primary grid contains 192×192 samples over a 2.4 mm square window, giving $\Delta x = 12.5\,\mu$m. The circular input aperture has radius 0.9 mm. All fields have power $P = 1$ W and wavelength $\lambda = 633$ nm. Propagation is through free space without a lens. The 13 observation planes span $z = 20$–80 mm at 5 mm intervals. Table 1 gives the input scales and phase coefficients.

The implementation checks finite fields, equal input power, retained spectral power of at least 0.995, and an edge-energy fraction no greater than 0.01. The edge region is the outer 12% of the grid on each side. The research proposal specified a stricter edge budget of 10^{-3}; this distinction matters for the Gaussian quartic field, whose maximum edge fraction is 1.39×10^{-3}. Passing the implementation's screening checks therefore does not imply that every condition in the proposal

Case	Input	s (μm)	α	Applied phase	Seed
C0001H001	Gaussian	—	—	quartic, $a_4 = 0.7$	101
C0001H002	Pearcey	200	—	quartic, $a_4 = 0.7$	202
C0001H003	Butterfly	200	—	quartic, $a_4 = 0.7$	303
C0001H004	Airy	100	0.1	quartic, $a_4 = 0.7$	404
C0001H005	Gaussian	—	—	chirp 0.7 + ripple 0.5	505

Table 1: Primary input fields and transformations. The Gaussian envelope radius is $w = 300\,\mu$m wherever it appears in the field definition. Seeds identify the separate random-phase diagnostics; the primary free-space comparisons have no random phase screen.

Input	Base η	Transformed η	Change	Screening
Gaussian	0.0520747	0.0512417	−1.60%	engine pass
Pearcey	0.0600749	0.0587485	−2.21%	engine pass
Butterfly	0.0621276	0.0610955	−1.66%	engine pass
Airy	0.0403029	0.0391722	−2.81%	retention/edge fail
Gaussian control	0.0520747	0.0369704	−29.00%	engine pass

Table 2: Median on-axis core-energy fractions before and after the phase transformations in Table 1. Change means $100(\eta_{\text{transformed}} - \eta_{\text{base}})/\eta_{\text{base}}$. Screening refers to the implementation thresholds in Section 4; the Airy values are retained as failed numerical diagnostics.

was met. Separate perturbation fields use independent Gaussian phase errors of rms 0.02 rad with random seed equal to the listed seed plus 17. This phase error is absent from the primary base and quartic fields.

For Gaussian and Pearcey inputs, further calculations use grids of 256 and 384 samples, a real-space Fresnel calculation at 384 samples, a 3.6 mm window with 576 samples, and 41 axial planes. The expanded window retains the 6.25 μm spacing of the 384-sample grid. A separate 640 nm diagnostic scales the stored principal launch phase by 633/640. An eight-level quantization diagnostic rounds the entire launch phase and is evaluated separately from the unquantized comparisons.

Three additional Gaussian phase families were searched with 32 attempts each, using the same power, aperture and propagation geometry. The bounded families contain quadratic phase, quadratic-plus-quartic phase, and a four-parameter phase. The recorded selection direction minimizes core efficiency. Their selected values consequently describe energy removal from the core, rather than optimized concentration.

5 Results and discussion

5.1 Response of the fixed core

The quartic phase reduces core efficiency for the Gaussian, Pearcey and Butterfly inputs (Table 2). Their unmodified core fractions differ because the input fields differ. The relevant comparison is the change within each family under the same phase transformation: −1.60%, −2.21% and −1.66%, respectively. These three changes have the same sign and lie within a range of 0.61 percentage points. The comparison is at one phase coefficient and supplies no statistical test of equality across families.

Figure 1 shows these paired core fractions together with the longitudinal marginal intensity of the quartic Gaussian field. The decrease in η measures redistribution outside a fixed core, not an equivalent decrease in total propagated power. For example, the Gaussian quartic field retains a minimum output power of 0.999997 W while its core fraction decreases from about 0.0521 to 0.0512.

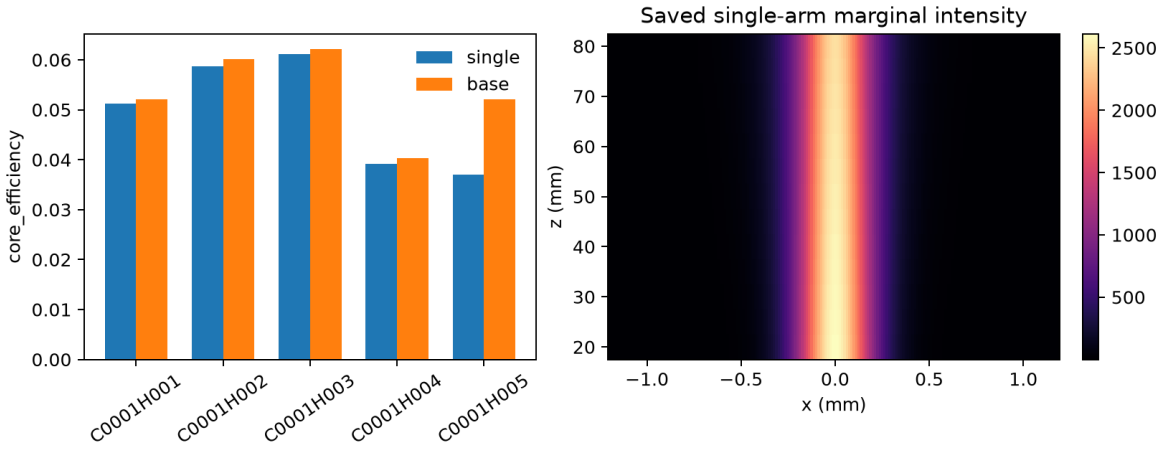


Figure 1: Left: transformed (single) and base core-energy fractions for the five cases in Table 1, in the same order. The Airy pair is a failed diagnostic. Right: longitudinal marginal intensity of the quartic Gaussian field, integrated over $|y| < 0.6\,\mathrm{mm}$. The color scale is in $\mathrm{W\,m^{-1}}$; the image shows a marginal profile rather than the circular-core integral used for η.

Calculation	Gaussian change (%)	Pearcey change (%)
Primary: 192^2, 2.4 mm	−1.600	−2.208
256^2, 2.4 mm	−1.612	−2.222
384^2, 2.4 mm	−1.601	−2.209
Direct Fresnel, 384^2	−1.601	−2.209
Expanded window: 576^2, 3.6 mm	−1.601	−2.209
41 axial planes, 384^2	−1.601	−2.184
Wavelength diagnostic: 640 nm	−1.685	−2.294

Table 3: Relative changes recomputed from the saved core fractions for each condition. Unless specified otherwise, the additional calculations use 384^2 samples, the 2.4 mm window, 13 axial planes and 633 nm. The wavelength row changes the physical condition as described in Section 4. No corresponding confirmation series is available for Butterfly.

The chirp-plus-ripple Gaussian comparison yields $\eta = 0.0369704$, a decrease of 29.0% from the same Gaussian base. This larger response establishes the scale of change produced by another specified radial mask. Because both its quadratic and ripple terms differ from the quartic-only case, it does not isolate which term causes the difference. Its radial symmetry also means that it is not the non-centrosymmetric random-phase control envisaged in the original proposal.

The Airy calculation loses appreciable power from the usable numerical representation. Its minimum output power is 0.9424 W and its maximum edge-energy fraction is 0.1105, compared with 0.0317 for the base Airy field. It fails both retention and edge checks. Its percentage change is therefore listed for completeness and excluded from the physical comparison of quartic responses.

5.2 Dependence on numerical sampling

Table 3 collects the additional Gaussian and Pearcey calculations. The mesh and window changes preserve the sign of both responses. Between the 192-, 256- and 384-sample grids, the Gaussian change ranges from −1.600% to −1.612% and the Pearcey change from −2.208% to −2.222%. These are variations of the paired relative changes, rather than uncertainty estimates for the absolute core fractions.

At 384^2 samples, the angular-spectrum and direct Fresnel calculations agree in the reported relative changes to the displayed three decimal places. Increasing the axial sampling to 41 planes

changes the Pearcey result from -2.209% to -2.184%, showing a small dependence on how the median samples the axial evolution. The 640 nm diagnostic retains the negative sign, although its change of wavelength and launch phase is a different physical condition.

The absolute core fractions have a larger sampling dependence than their ratios. For the Gaussian input, the base value is 0.0520747 at 192 samples, 0.0531806 at 256 samples and 0.0523381 at 384 samples. Integrating a hard circular boundary on different grids can change the included sample set. The close paired changes therefore support numerical repeatability of the relative response under these tests, without supplying a complete error bound on η. The proposal's 5% convergence tolerance is an acceptance threshold, not a measured error bar or a detector resolution. It cannot by itself establish that a 2% relative effect is unresolved.

Phase quantization behaves differently. The eight-level diagnostics give changes of $+8.41\%$ for Gaussian and -10.76% for Pearcey, and both fail the retained-spectrum check. The Pearcey diagnostic also fails the integral-overlap check. They are excluded from the unquantized response and from any claim of tolerance to quantization.

5.3 Meaning of the quadratic comparisons

The selected quadratic Gaussian mask has $q_2 = 1.2$ and $\eta = 0.0408236$. The selected quadratic-plus-quartic mask has $q_2 = 1.2$, $q_4 = 0.7$ and $\eta = 0.0403671$. The four-parameter search returns the same point as the latter. Thus, adding the quartic term at that selected quadratic coefficient decreases the core fraction by a further 1.12%. These values belong to searches for lower core efficiency; they do not establish an improvement in energy concentration or a global optimum.

None of those masks was constructed from Eq. (8). Consequently they cannot supply η_{equiv} in Eq. (9). The observed 1.6–2.2% quartic-versus-base changes also cannot be compared directly with the proposed threshold $R > 0.25$, which has a different denominator. The available simulations answer how each field responds to the specified quartic phase, while the moment-equivalence question still requires its own quadratic comparison.

6 Physical interpretation and scope

The radial phase gradients make the distinction between the two masks explicit. For $\phi_4 = a_4(r/w)^4$, the added gradient is $\partial_r\phi_4 = 4a_4r^3/w^4$; for $\phi_2 = a_2(r/w)^2$, it is $2a_2r/w^2$. The positive quartic coefficient therefore adds an outward phase gradient whose magnitude grows more rapidly in the outer field. The observed loss of energy from the on-axis core is compatible with this redistribution. The accumulated core energy also depends on diffraction and the input's existing phase, so the gradient alone does not determine its magnitude.

This distinction matters for interpreting similar numerical changes. Gaussian and caustic fields need not share an equivalent propagation law merely because the same mask changes their median core energies in the same direction. The three primary results are specific to $a_4 = 0.7$, the 50 μm core and the 20–80 mm axial interval. Gaussian and Pearcey have additional sampling checks; Butterfly has one primary calculation. No parameter sweep or repeated-run distribution establishes equality across the families.

The missing quadratic comparison limits the proposed mechanism test, rather than erasing the measured quartic response. Evaluating M on each frozen input and propagating the resulting quadratic field would make R calculable, subject to a nonzero denominator and an error analysis for the difference. Until that comparison is available, the common sign of the measured changes describes the response of these fields in this geometry.

7 Conclusions

A radial quartic phase with $a_4 = 0.7$ decreases the median energy fraction within a 50 μm on-axis core by 1.60% for Gaussian, 2.21% for Pearcey and 1.66% for Butterfly fields over 20–80 mm of free-space propagation. The Gaussian and Pearcey relative changes remain close under the reported mesh, window and independent-propagator checks. The chirp-plus-ripple Gaussian mask produces a larger 29.0% decrease, while the Airy and quantized calculations fail numerical screening. The results describe a small reduction of core energy at one quartic coefficient. They leave the proposed intensity-moment reduction to quadratic phase untested.

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Disclosures

The authors declare no financial conflicts of interest. The study uses numerical propagation.

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